

Hawkeye EQMS · Supplier Pre-Qualification · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

Document version	v1.1
Generated on	2026-04-26T00:06:31.924Z
Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

Contents

- 1. Module overview
- 2. Standard pharma EQMS vs Hawkeye — gap analysis
- 3. Personas + role mapping
- 4. Feature catalogue (click-by-click)
- 5. Full lifecycle walkthrough — one transaction, end-to-end
- 6. AI assistance map
- 7. Regulator-trace matrix
- 8. Test results matrix
- 9. Known gaps + roadmap

1. Module overview

Module	Supplier Pre-Qualification
Pharma purpose	Screen a new supplier before scheduling a full audit. Capture initial risk band, regulatory standards, product categories, checklist compliance. Output: APPROVED / CONDITIONALLY_APPROVED / REJECTED with validity window.
Compliance basis	ICH Q10 §2.7 · 21 CFR 211.84 (component qualification) · EU GMP Chapter 7 · ISO 13485 §7.4
Hawkeye routes	/supplier-prequalification (frontend) · /api/supplier-prequalifications (backend)
Hawkeye model file	backend/src/models/SupplierPreQualificationModel.js
Tenant module flag	modules.SUPPLIER_QUALITY

Pre-quals follow DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED / CONDITIONALLY_APPROVED / REJECTED. An APPROVED PQ becomes the basis for scheduling a full audit. CONDITIONALLY_APPROVED carries explicit conditions tracked separately. EXPIRED is auto-set when validUntil passes.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. MET = parity, PARTIAL = ships but with caveats, GAP = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
Auto-numbered (PQ-YYYY-NNNN) GMP	pqNumber pre-save generator.	MET	
Risk band capture (LOW / MEDIUM / HIGH) ICH Q9	initialRiskBand enum.	MET	
Regulatory standards + product categories ICH Q10 §2.7	regulatoryStandards[] + productCategories[].	MET	
Checklist with criterion + result + notes 21 CFR 211.84	checklist[] subdocument.	MET	
Decision (APPROVED / CONDITIONALLY_APPROVED / REJECTED) with rationale + validUntil ICH Q10 §2.7	/decision endpoint with decision + decisionNotes + conditions[] + validUntil.	MET	
Auto-EXPIRE on validUntil —	Vercel cron (03:30 UTC daily) calls /api/quality/scan-expirations → flips APPROVED/CONDITIONALLY_APPROVED PQs past validUntil to EXPIRED.	MET	
Escalate APPROVED PQ to full audit request ICH Q10 §2.7	auditRequestId field links the PQ to the spawned audit.	MET	
Public-data enrichment (FDA warning letters / 483s / import alerts) FDA AI guidance	Wave audit-agents · Supplier-Intel agent (POST /api/ai/audit-agents/supplier-intel) returns openFDA + EMA + WHO PQ signals.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Program Mgr Priya buyer · PQ owner	<code>audit.program@novex-pharma.demo</code>	Initiates PQ for new suppliers, drives review, escalates to audit.	DRAFT SUBMITTED UNDER_REVIEW

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Supplier (external) supplier	(supplier-side login)	Fills the questionnaire response.	DRAFT (external)
Audit Lead Maria auditor · approver	audit.lead@novex-pharma.demo	Reviews checklist, signs off APPROVED / CONDITIONAL / REJECTED.	UNDER_REVIEW APPROVED CONDITIONALLY_APPROVED REJECTED
VP Quality Elena tenant_admin	vp.quality@novex-pharma.demo	Final approval on HIGH-risk supplier prequals.	APPROVED

4. Feature catalogue (click-by-click)

4.1 PQ register

What it does	Lists all pre-quals with status + risk band + decision.
Menu / route	/supplier-prequalification
Available to roles	any tenant viewer
Backend API	GET /api/supplier-prequalifications

CLICK-BY-CLICK

- NAVIGATE** Click 'Suppliers' → 'Pre-qualification'
→ Page renders

Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Supplier Prequalification

Supplier Pre-Qualification

Desk review & initial risk screening before a full Internal GMP Audit is initiated

+ New PQ

Filter by status

[PQ-2026-0003] API supplier qualification for Novexolimus drug substance.
Status: APPROVED · Created 4/24/2026

[PQ-2026-0002] API supplier qualification for Novexolimus drug substance.
Status: APPROVED · Created 4/24/2026

[PQ-2026-0001] API supplier qualification for Novexolimus drug substance.
Status: APPROVED · Created 4/24/2026

state-screens/supplier-prequal-list.png

4.2 + Start Pre-Qual dialog

What it does	Initiate a new PQ.
Menu / route	/supplier-prequalification · top-right '+' Start Pre-Qual'
Available to roles	supplier · buyer · admin
Backend API	POST /api/supplier-prequalifications

CLICK-BY-CLICK

- CLICK** Click '+ Start Pre-Qual'
→ Dialog opens
- TYPE** Fill supplierName
→ required
- TYPE** Fill scope (1 sentence)
→ required
- CLICK** Pick initialRiskBand (LOW / MEDIUM / HIGH)
→ required
- TYPE** Fill regulatoryStandards[] (e.g. ICH Q7, 21 CFR 211)
- TYPE** Fill productCategories[] (e.g. API, Excipient)
- CLICK** Click 'Save'
→ Row · status=DRAFT · pqNumber=PQ-YYYY-NNNN

FIELDS CAPTURED

Field	Required	Validation / values	Notes
supplierName	Yes	string	
scope	Yes	string	
initialRiskBand	Yes	LOW MEDIUM HIGH	
regulatoryStandards	No	string[]	
productCategories	No	string[]	

4.3 Submit (DRAFT → SUBMITTED)

What it does	Supplier submits the PQ for buyer review.
Menu / route	PQ row · 'Submit' button
Available to roles	supplier
Backend API	PUT /api/supplier-prequalifications/:id (status=SUBMITTED)

CLICK-BY-CLICK

- CLICK** Click 'Submit'
→ Status flips

4.4 Begin review (SUBMITTED → UNDER_REVIEW)

What it does	Buyer accepts the PQ into the queue.
Menu / route	PQ row · 'Begin Review' button
Available to roles	buyer
Backend API	PUT /api/supplier-prequalifications/:id (status=UNDER_REVIEW)

CLICK-BY-CLICK

1. **CLICK** Click 'Begin Review'
→ Status flips

4.5 Decision (UNDER_REVIEW → APPROVED/CONDITIONAL/REJECTED)

What it does	Auditor signs off with decision + validUntil + (optional) conditions.
Menu / route	PQ row · 'Decision' button
Available to roles	auditor · tenant_admin
Backend API	POST /api/supplier-prequalifications/:id/decision

CLICK-BY-CLICK

1. **CLICK** Click 'Decision'
→ Drawer with decision picker
2. **CLICK** Pick decision (APPROVED / CONDITIONALLY_APPROVED / REJECTED)
→ required
3. **TYPE** Fill decisionNotes (rationale)
→ required
4. **CLICK** Pick validUntil (e.g. +2 years)
→ required for APPROVED / CONDITIONAL
5. **TYPE** (If CONDITIONAL) Fill conditions[]
6. **CLICK** Click 'Submit'
→ Status flips · decisionAt set

4.6 AI · Supplier Intel agent

What it does	Pull public FDA + EMA + WHO PQ signals about the supplier.
Menu / route	PQ detail · 'Check public signals' button
Available to roles	any tenant user
Backend API	POST /api/ai/audit-agents/supplier-intel
AI assistance	AI Cross-product: openFDA + FDA warning letters + EMA EudraGMDP + WHO PQ. Verdict = known_tenant / public_only / ambiguous / unknown.

CLICK-BY-CLICK

1. **CLICK** Click 'Check public signals'
→ Drawer opens

2. **WAIT** Wait ~3-5 s for the public-data fusion
→ Returns FDA ANDAs + warning letters + import alerts; verdict shown

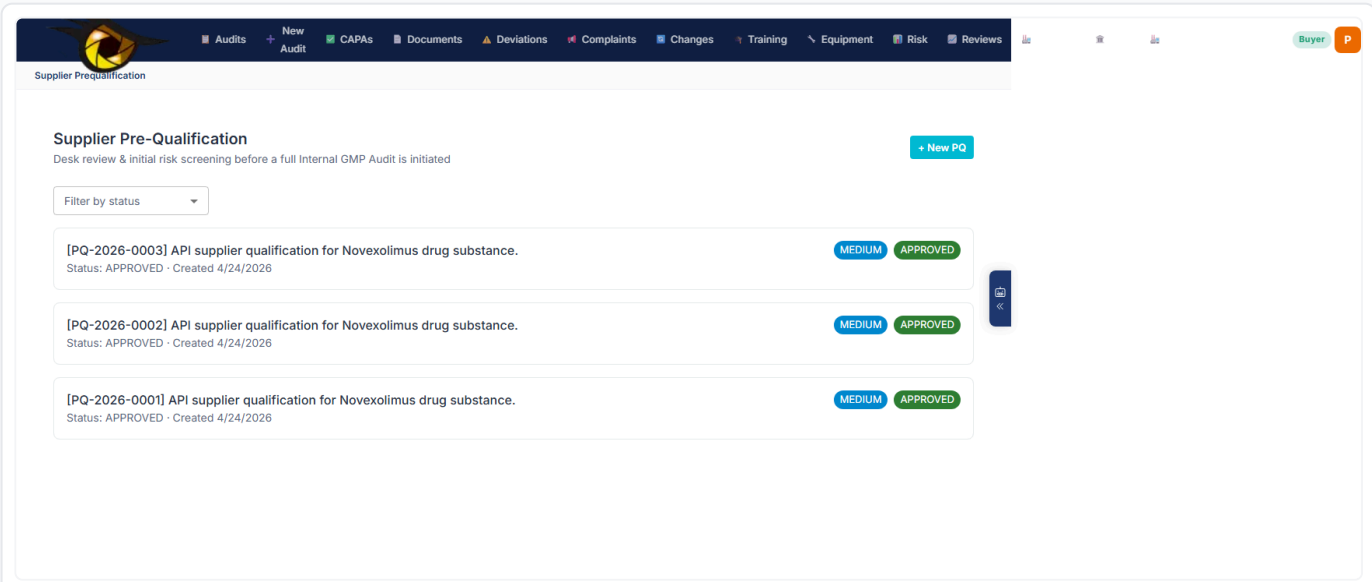
5. Full lifecycle walkthrough — one transaction, end-to-end

One PQ walked from start to APPROVED.

#	Persona	State transition	Action	API	Observed	Result
1	Priya Audit Program Mgr	— → DRAFT	+ Start Pre-Qual → supplierName='E2E Lifecycle Supplier', scope, initialRiskBand=MEDIUM, regulatoryStandards= [ICH Q7, 21 CFR 211], productCategories=[API, Raw material]	POST /api/supplier- prequalifications	pqNumber=PQ- 2026-NNNN · status=DRAFT	PASS

Expected DB state after this step:

```
supplier-prequalifications { _id, pqNumber, supplierName, scope, initialRiskBand: 'MEDIUM', regulatoryStandards,  
productCategories, status: 'DRAFT', initiatedBy }
```



The screenshot shows a web application interface for 'Supplier Pre-Qualification'. At the top, there's a navigation bar with various icons and a 'Buyer' button. Below the navigation bar, the main heading is 'Supplier Pre-Qualification' with a subtitle 'Desk review & initial risk screening before a full Internal GMP Audit is initiated'. A '+ New PQ' button is visible. Below this, there's a 'Filter by status' dropdown menu. The main content area displays a list of three PQs, each with a 'MEDIUM' risk band and an 'APPROVED' status. The PQs are for 'Novexolimus drug substance'. The interface is clean and modern, with a light blue and white color scheme.

state-screens/supplier-prequal-list.png

2	Priya Buyer	DRAFT → SUBMITTED	PUT status=SUBMITTED	PUT /api/supplier- prequalifications/:id	Status flips	PASS
3	Priya Buyer	SUBMITTED → UNDER_REVIEW	PUT status=UNDER_REVIEW	PUT /api/supplier- prequalifications/:id	Status flips	PASS

#	Persona	State transition	Action	API	Observed	Result
4	Maria Lead Auditor	UNDER_REVIEW → APPROVED	POST decision · decision=APPROVED · decisionNotes · validUntil=+2y	POST /api/supplier- prequalifications/:id/decision	Status=APPROVED · decisionAt set	PASS
Expected DB state after this step:						
supplier-prequalifications { status: 'APPROVED', decision: 'APPROVED', decisionNotes, validUntil, decidedBy: maria._id, decisionAt }						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Supplier-Intel Agent (audit-agents)	DRAFT UNDER_REVIEW	POST /api/ai/audit-agents/supplier-intel	PQ detail · 'Check public signals'	openFDA + FDA warning letters + import alerts + EMA EudraGMDP + WHO PQ; verdict = known_tenant / public_only / ambiguous / unknown	Public-data fusion (no LLM for ranking; LLM for narrative)

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
DRAFT	ICH Q10 §2.7	supplierName + scope + initialRiskBand + regulatoryStandards + productCategories	supplier-prequalifications
UNDER_REVIEW	21 CFR 211.84(d)(1)	checklist[] populated	supplier-prequalifications
APPROVED	21 CFR 211.84 ICH Q10 §2.7	decision=APPROVED + decisionNotes + decidedBy + decisionAt + validUntil	supplier-prequalifications

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · supplier-prequal	DRAFT → SUBMITTED → UNDER_REVIEW → APPROVED	PASS	4/4 PASS · eqms-test-results-v2.pdf

9. Known gaps + roadmap

- **Auto-EXPIRE scheduler** — Cron flips status to EXPIRED when validUntil passes. **HIGH**
- **Auto-create audit request from APPROVED PQ** — One-click 'Schedule Full Audit' button on APPROVED PQ. **HIGH**
- **AI Supplier-Intel button on the +Start dialog** — Pre-fill the PQ with public signals before submission. **MEDIUM**
- **Conditions tracker module** — CONDITIONALLY_APPROVED PQs need ongoing tracking; today conditions are free-text. **MEDIUM**